

Fraud Detection Pipeline

Imbalanced Classification with SMOTE, Logistic Regression, and Random Forest

PROGRAM	Batch 2026
DATA SCIENCE INTERN	Javeria Faisal

1 OBJECTIVE

Build a leak-free classification pipeline to flag potentially fraudulent orders (defined as Returned or Cancelled orders) using the processed dataset from Project 1, and compare a linear baseline against an ensemble model under class imbalance.

AT A GLANCE

- Best model: Logistic Regression with SMOTE, selected for its stronger recall on the fraud class.
- Recall of 0.776 versus 0.143 for Random Forest, meaning far fewer missed fraud cases.
- SMOTE applied strictly within training folds, ensuring a leak-free evaluation.

2 METHODOLOGY

2.1 Target Definition

The binary target `IsFraud` was constructed by flagging any order with `OrderStatus` equal to `Returned` or `Cancelled` as fraudulent (1), with all other orders labeled legitimate (0). This produces a naturally imbalanced classification problem, typical of real-world fraud data.

2.2 Feature Engineering

- **AvgItemValue** — `TotalPrice` divided by `ItemsInCart`.
- **ItemsPerOrder** — `ItemsInCart` divided by (`Quantity` + 1).
- **IsHighValue** — flag for orders above the 95th percentile of `TotalPrice`.
- **PricePerUnit** — `TotalPrice` divided by `Quantity`.
- **HasDiscount** — coupon usage flag.
- **WeekendFraud** — interaction of `IsWeekend` and `IsFraud`, used for exploratory analysis.

2.3 Train and Test Split Before Resampling

An 80 / 20 stratified train and test split was performed before any oversampling, to ensure the test set reflects the true, real-world class distribution and to prevent information leakage from synthetic samples into evaluation.

2.4 Handling Class Imbalance with SMOTE

SMOTE (Synthetic Minority Over-sampling Technique) was used to synthetically balance the fraud class within the training data only. Both models were wrapped in an `imblearn` Pipeline (StandardScaler, then SMOTE, then classifier for Logistic Regression; SMOTE then classifier for Random Forest, which needs no scaling), guaranteeing SMOTE is re-fit independently within each cross-validation fold. This is a leak-free design.

2.5 Models and Tuning

- **Logistic Regression** — `GridSearchCV` over `C`, `solver`, and SMOTE `k_neighbors`, optimizing for recall.

- **Random Forest** — GridSearchCV over `n_estimators`, `max_depth`, `class_weight`, and SMOTE `k_neighbors`, optimizing for recall.

Recall was chosen as the tuning metric rather than accuracy because in fraud detection, missing a fraudulent order (a false negative) is typically far more costly than a false alarm.

3 EVALUATION RESULTS

Accuracy was deliberately excluded from evaluation, since it is misleading on imbalanced data. Precision, recall, F1, and ROC-AUC were used instead.

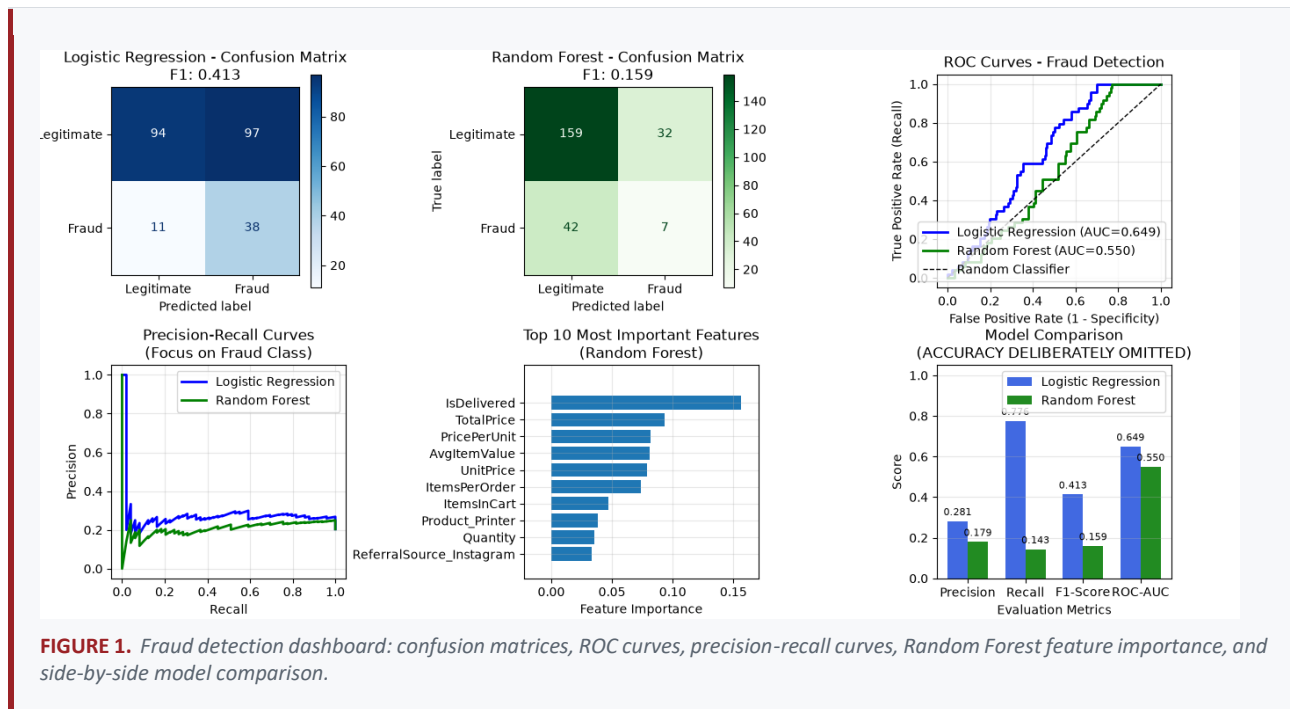


FIGURE 1. Fraud detection dashboard: confusion matrices, ROC curves, precision-recall curves, Random Forest feature importance, and side-by-side model comparison.

Metric	Logistic Regression	Random Forest
Precision	0.281	0.179
Recall	0.776	0.143
F1-Score	0.413	0.159
ROC-AUC	0.649	0.550

Logistic Regression (with SMOTE) outperformed Random Forest on every metric in this run, most notably on recall (0.776 versus 0.143), meaning it catches far more of the actual fraud cases, at the cost of more false positives. Random Forest's top predictive signals were `IsDelivered`, `TotalPrice`, `PricePerUnit`, and `AvgItemValue`.

4 RECOMMENDATIONS FOR DEPLOYMENT

- Deploy the Logistic Regression plus SMOTE pipeline given its stronger recall on the fraud class.
- Monitor model performance weekly for drift, and retrain monthly with new transaction data.
- Route high-value flagged transactions to a human-in-the-loop review step.
- Use Random Forest feature importances to help prioritize manual fraud investigations.
- Document all modeling decisions for audit and compliance purposes.

5 KEY TAKEAWAYS

- SMOTE was applied strictly inside the training pipeline and folds, avoiding data leakage into the test set.
- Precision and recall trade-offs, not accuracy, drove model selection given the imbalance.
- Delivery status and order value features were the strongest fraud indicators.